

Nitish Ravisankar Raveendran

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github.com/Nitish05

Education

- University of Maryland, College Park**, M.Eng Robotics Aug 2023 – May 2025
- GPA: 3.79/4.0
 - **Coursework:** Machine Learning, AI in Robotics, Sensor Integration, Computer Vision, Sensor Fusion, SLAM
- Anna University, PEC, Chennai**, B.E. Electronics and Communications Aug 2018 – May 2022
- GPA: 8.94/10

Experience

- Software Engineer**, Cognizant, Chennai Feb 2022 – Aug 2023
- Developed and deployed full-stack features using React and Node.js, improving system responsiveness by 15%.
 - Enhanced application security by implementing OAuth-based authentication, reducing vulnerabilities.
- Club President**, Robotics Club, PEC Chennai, March 2019 – Aug 2022
- Championed club initiatives that increased membership and orchestrated multiple innovative projects and events, engaging a large number of students.

Projects

- Bipedal Locomotion Using Reinforcement Learning** github.com/Nitish05/walker
- Analyzed and optimized model training by implementing and comparing TRPO, PPO, and TQC algorithms to generate adaptive bipedal gaits. Reduced training convergence time by 35% and improved sample efficiency, demonstrating a strong grasp of how architectural choices and optimization techniques impact training performance in complex sequential decision-making problems.
 - Tools Used: Python, PyTorch, OpenAI Gym, MuJoCo, ROS 2
- Reinforcement Learning for Pursuit-Evasion** github.com/Nitish05/p&e
- Developed a multi-agent DQN framework in ROS2 & Gazebo to investigate emergent behaviors and convergence challenges in a distributed agent system. This work provides insights into training stability and parameter tuning for complex and interactive artificial intelligence models analogous to real-world driving scenarios.
 - Tools Used: Python, PyTorch, ROS 2, Gazebo, Docker
- StereoSight: Local Planner for Drones** github.com/Nitish05/SS
- Engineered a real-time perception and planning pipeline in ROS 2, processing stereo camera depth maps into occupancy grids for autonomous navigation. Optimized Dijkstra algorithm for efficient waypoint computation, crucial for low-latency decision making in production systems.
 - Tools Used: C++, Python, ROS 2, OpenCV, PCL
- TurtleBot3 Vision-Based Path Following** github.com/Nitish05/TB
- Designed a robust vision-based perception and control pipeline in ROS2 and OpenCV to enhance autonomous navigation. This system forms a foundational block for end-to-end models, successfully navigating complex paths with varied conditions, increasing route success rates by 40% while maintaining 30 FPS processing.
 - Tools Used: C++, Python, OpenCV, ROS2, Linux

Technologies

Languages: Python, C++, C#, JavaScript, SQL

Technologies: ROS 2, Gazebo, Docker, PyTorch, OpenAI Gym, MuJoCo, OpenCV, PCL, Git, Linux, Isaac Sim, React, Node.JS